<init>

Get the serial port parameters from the parameter server

Is the parameter server up? (are we running from a launch file?)

No

Use default parameters

Yes

Start listening on serial port

<Spin until bytes are available>

Read all available bytes from the port into a string

Find the first "$GPGGA"

Do we have GPS lock?

No

Print error message

Yes

Convert latitude string to degrees

Convert longitude string to degrees

Flip the sign of longitude depending on West/East reading

Convert to UTM measurement

Publish GPS Message

Advance buffer by amount of data in GPGGA message